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Retinoic acid amplifies the host immune response to LPS through increased T lymphocytes number and LPS binding protein expression

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Abstract

Vitamin A deficiency is associated with increased susceptibility to infection but the effects of Vitamin A supplementation on host response to pathogens are controversial. This study investigated the mechanisms by which all-trans retinoic acid (atRA) modulates the host immune response in an experimental model of Vitamin A supplementation before and after challenge with LPS in rats. We show here that a supplementation with five daily injections of 10 mg/kg atRA increased the number of T lymphocytes in the peripheral blood. In addition, we show that atRA increased the expression of the LPS binding protein (LBP), a component of the LPS recognition system. The retinoic acid receptor (RAR)α agonist Ro 4060-55 but not the pan-retinoid X receptors (RXRs) agonist Ro 2573-86 mimicked the effects of atRA on LBP expression suggesting that atRA enhances LBP expression through a RARα-mediated pathway. In order to investigate the significance of increased LBP expression we challenged atRA-supplemented rats with the Gram-positive bacteria *Listeria monocytogenes* (LM) that activates the immune response independently from LBP. In sharp contrast to our previous observations that atRA supplementation enhances IFN-γ expression and NOS2 pathway activation in LPS-challenged rats [Devaux, Y., Grosjean, S., Seguin, C., David, C., Dousset, B., Zannad, F., Meistelman, C., de Talancé, N., Mertes, P.M., Ungureanu-Longrois, D., 2000. Retinoic acid and host–pathogen interactions: effects on inducible nitric oxide synthase in vivo. Am. J. Physiol. 279, E1045–E1053], atRA did not increase the LM-induced IFN-γ expression and NOS2 pathway activation. Overall, these data demonstrate that although atRA induces a "priming" of the immune system characterized by increased T lymphocytes number and LBP expression, the profile of the immune response depends on the inflammatory/infectious stimulus. These results could explain why Vitamin A supplementation could have beneficial/neutral or deleterious effects according to the identity of the infectious pathogen.
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Keywords: Vitamin A; Lipopolysaccharide; *Listeria monocytogenes*; T cells; LPS binding protein

1. Introduction

Vitamin A deficiency has been convincingly shown to be associated with increased susceptibility to infection. Under certain circumstances, supplementation with Vitamin A has been reported to decrease the incidence and severity of infectious disease, which gained its name as the "anti-infective" vitamin (Semba, 1999; Villamor and Fawzi, 2000). While the majority of trials have reported beneficial effects of Vitamin A supplementation, others have reported that Vitamin A supplementation had neutral (Barreto et al., 1994; Binka et al., 1995; Fawzi et al., 1998; Nacul et al., 1998) or even, in some circumstances, deleterious effects (Stansfield et al., 1993; Stephensen et al., 1998) on the incidence or course of infection. The beneficial versus detrimental or neutral effects of Vitamin A supplementation probably depend on factors, such as the host Vitamin A status before the supplementation (Vitamin A deficient or replete subjects), the hormonal status of the host, the timing of supplementation

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(prophylaxis or therapy of infectious diseases), the chemical identity of the Vitamin A derivative, the endpoints used to assess the effects of supplementation (Villamor and Fawzi, 2000; Rosales, 2002; Fawzi et al., 2003) and also the pathogens and the organs involved. In addition, several regimens of supplementation have been reported, such as unique large doses, large but spaced doses and frequently repeated small doses, the last being probably the most efficient (Villamor and Fawzi, 2000).

The mechanisms that mediate the "anti-infectious" effects of Vitamin A supplementation are thought to be related to changes in the host inflammatory and immune (both innate and adaptive) responses thus suggesting a stereotyped immune modulating effect of Vitamin A supplementation. Among the effectors of the innate immune response, Vitamin A and its active metabolite, all-trans retinoic acid (atRA), modulate the expression and/or activity of several pathways involved in the host response to pathogens, such as the enzymes responsible for prostanoids biosynthesis, i.e. phospholipases A2 (Hope et al., 1990) and cyclooxygenases (Nüsing et al., 1995; Hill et al., 1996; Devaux et al., 2001), the inducible nitric oxide synthase (NOS2) (Devaux et al., 2000; Austenaa and Ross, 2001) as well as several cell types, such as neutrophils, monocytes/macrophages and natural killer (NK) cells (Semba, 1998; Erickson et al., 2000). Vitamin A also modulates the adaptive immune response and specifically the biology of dendritic cells (Geissmann et al., 2003), the antigen-mediated activation of T lymphocytes (Friedman et al., 1993; Iwata et al., 2003; Ertesvag et al., 2002) as well as B cells differentiation and immunoglobulin production (Blomhoff et al., 1992).

In previous studies we investigated the effects of atRA on several components of the innate immune response, such as NOS2 pathway, an important actor of the host response to infectious stimuli. We demonstrated that atRA supplementation in rats in vivo enhanced the LPS-induced activation of the NOS2 pathway (Devaux et al., 2000), in sharp contrast to the in vitro data reported by several groups (Hirokawa et al., 1994; Mehta et al., 1994; Datta and Lianos, 1999; Datta et al., 2001; Motomura et al., 1997) including ours (Grosjean et al., 2001). To explain the differences between in vivo and in vitro effects of atRA supplementation on the NOS2 pathway, we hypothesized that atRA may amplify the host response in vivo through mechanisms absent in vitro. We report here that atRA increased T lymphocytes number in the peripheral blood. In addition, we observed that atRA enhanced the expression of the LPS binding protein (LBP), a component of the LPS recognition system. Our results suggest that enhancement of LBP expression by atRA depends on retinoic acid receptor (RAR)α, since the RARα agonist Ro-4060-55 but not the pan-retinoid X receptors (pan-RXRs) agonist Ro 2573-86 mimicked the effects of atRA on LBP expression.

We attempted to investigate the significance of the atRA-mediated enhancement of LBP expression by challenging atRA-supplemented rats with the Gram-positive bacteria *Listeria monocytogenes* (LM) that activates the host immune response through mechanisms independent from LBP. In sharp contrast to our previously published results showing that atRA increased the LPS-induced IFN-γ expression and NOS2 pathway activation, atRA did not modify the activation of these two components of innate immunity when the infectious agent was LM.

2. Materials and methods

2.1. Reagents

All chemicals and reagents were purchased from Sigma (Saint Quentin Fallavier, France) unless specified otherwise. *Salmonella typhimurium* LPS (Lot 96H4021) was dissolved in 0.9% (w/v) NaCl at a concentration of 4 mg/mL and stored at 4 °C. The previously characterized (Dramsi et al., 1997) LM strain EGD-E was kindly provided by Dr. Pascale Cossart. Aliquots of 10^9 colony forming units (CFU) LM in 0.9% (w/v) NaCl containing 20% glycerol were prepared and stored at −80 °C. Preliminary experiments were performed to document that bacteria viability and number were not significantly altered by storage. The RARα agonist Ro 4060-55 and the pan-RXRs agonist Ro 2573-86 were provided by Dr. R. Klaus (Hoffmann-La Roche, Basel, Switzerland). atRA, RARα and RXRs agonists were dissolved in 5% (v/v) DMSO and cold first pressure olive oil at a concentration of 10 mg/mL; aliquots were stored protected from light at −20 °C. Experiments involving retinoids were conducted under reduced light conditions. LPS and retinoids were administered i.p. in 500 μL volumes.

2.2. Experimental protocols

Male Wistar Kyoto rats (250–350 g) were housed and treated in accordance with the NIH Guide for the Care and Use of Laboratory Animals [1996 (7th ed.) Washington, DC: National Academy Press, aka National Research Council Guide].

2.2.1. Experiments with lipopolysaccharide

Rats were randomly divided into six groups ($n = 10$ in each group) as follows. For 5 consecutive days, rats received daily i.p. injections (10 mg/kg b.w.) of atRA (atRA and atRA + LPS groups), Ro-4060-55 (RARα group) or Ro 2573-86 (RXRs group). Rats from LPS and control groups received olive oil plus 5% DMSO in the same conditions. On day 5, rats from LPS and atRA + LPS groups were injected i.p. with LPS (4 mg/kg b.w.) and rats from control, atRA, RARα and RXRs groups with the same volume (i.e. 500 μL) of 0.9% (w/v) NaCl. Six hours thereafter, rats were anesthetized with 100 mg sodium thiopental (Nesdonal®, Rhône Poulenc Rorer, Paris, France) i.p., and the thorax and abdomen were dissected. Whole blood samples were recovered by cardiac puncture. Four milliliters were collected into heparinized tubes (Vacutainer®, Becton Dickinson, Meylan, France), centrifuged at 600 × *g* for 10 min and the plasma was stored at −70 °C. Tissue samples from liver, kidney and spleen were excised, rapidly rinsed in ice-cold saline, frozen in liquid nitrogen and stored at −70 °C until analysis.

2.2.2. Experiments with the facultative intracellular Gram-positive bacteria L. monocytogenes

Rats were divided in two groups: rats from atRA + LM group ($n = 5$) received five daily i.p. injections of atRA (10 mg/kg b.w.), whereas rats from LM group ($n = 5$) received olive oil plus 5% (v/v) DMSO under the same conditions. On day 5, rats from both groups were infected with 10^9 CFU LM i.p. and were sacrificed 24 h later. Rats were anesthetized, the thorax and abdomen were dissected, and samples from liver and spleen were recovered as described above. Additional samples from liver and spleen were harvested and freshly assayed for bacterial load. The dose of 10^9 CFU LM given intraperitoneally corresponds to the LD_{50} (Peress et al., 1985).

2.3. Measurement of bacterial load

After weighing, liver and spleen samples were homogenized in 2 mL 0.9% (w/v) NaCl with a potter, 10 μL aliquots were plated onto blood-gelose and CFU were counted after 24 h of growth at 37 °C. Each sample was assayed in triplicate, undiluted and diluted 10-fold. Results are expressed in CFU/g tissue.

2.4. Analysis of blood cells by flow cytometry

Differential white blood cells counts were obtained using a hematological counter (H2 Technicon, Bayer, France). Peripheral blood cells were double-stained with mAb directly conjugated either to fluoresceine isothiocyanate (FITC) or phycoerythrin (PE). This assay was performed in 100 μL samples of whole blood with 10 μL of mAb solution for 30 min at room temperature and in the dark. Two-color staining was performed to determine the number of T helper (Th) lymphocytes ($CD5^+/CD4^+$) and NK cells ($CD5^-/CD161^+$). All mAb were mouse anti-rat IgG1 and were purchased from Serotec Ltd. (Oxford, UK): CD5 mAb (clone OX-19), CD4 mAb (clone W3/25) and CD161 mAb (clone 10/78). The mAb IgG1-FITC and IgG1-PE (clone F8-11-13, Serotec) served as isotypic controls. Red blood cells were then lysed with IOTest® three lysing solution (Beckman Coulter, Marseille, France) for 10 min at room temperature. The samples were washed, centrifuged at 300 × *g* for 5 min, and suspended in CellWash® (Becton Dickinson, Meylan, France). Samples were analyzed on a FACScalibur flow cytometer (Becton Dickinson) and data were acquired and analyzed with CellQuest® software. The lymphocytes were identified by their low forward and side scatters and gated. Flow cytometry analysis was performed on 10,000 events.

2.5. Analysis of mRNA expression by semi quantitative RT-PCR

Messenger RNA expression of the housekeeping gene β-actin, IL-2, IFN-γ, NOS2 and several components of the host LPS recognition system (TLR4, CD14, MD-2 and LPS binding protein) was assessed by semi quantitative RT-PCR. The amplimer sequences, Genbank accession numbers and lengths of the expected PCR products are presented in Table 1. Amplification cycle numbers and annealing temperatures were optimized for each amplimer pair. Preliminary experiments were performed to document that the PCR was completed during the exponential phase of amplification and that the amplification was linear. The identity of the PCR products was confirmed by sequencing using ABI PRISM™ Dye Terminator Cycle sequencing Ready Reaction Kit and ABI PRISM™ 310 (PE Applied Biosystems). The PCR products were separated on a 2% agarose gel and densitometric analysis of the resulting bands was performed with a Bio-Rad Gel Doc 1000® (Bio-Rad Laboratories). Results were expressed as a ratio of the optical density of the band of the PCR product of interest to that of β-actin.

2.6. Measurement of IL-2 and LBP concentration

Concentration of IL-2 was determined in the plasma and liver homogenates by ELISA using a commercially available kit following the manufacturer's instructions (R&D Systems Europe, Lille, France). LBP plasma concentration was measured by ELISA using a kit from HyCult Biotechnology b.v. (Uden, The Netherlands).

2.7. Statistical analysis

Statistical analysis was performed using the StatView IV software (Abacus Concepts Inc., Berkeley, CA, USA). Results are expressed as mean ± S.E.M. Comparisons between two groups were performed with the Mann–Whitney test. A p-value <0.05 was considered statistically significant.

3. Results

3.1. Effect of supplementation with atRA on peripheral blood count and T lymphocytes subsets

In the peripheral blood of rats from all groups, monocytes and eosinophils populations corresponded to 1–4% of total leukocytes and atRA did not modify these percentages (data not shown). Supplementation with atRA increased the number of circulating leukocytes, neutrophils and lymphocytes in the peripheral blood compared with the control group (Fig. 1A). Since we previously reported that atRA supplementation in rats resulted in increased LPS-induced IFN-γ expression (Devaux et al., 2000), we sought to determine which cell population could be responsible for this effect. T lymphocytes and natural killer cells are the main producers of IFN-γ. We therefore performed flow cytometry experiments using markers specific for natural killer cells (CD161) and T lymphocytes (CD5). These experiments revealed that atRA increased the number of circulating $CD5^+$ cells from 3.4 ± 0.4 to $5.7 \pm 0.3 \times 10^9$ cells/L blood ($p = 0.0012$), but not the number of $CD5^-$ cells (Fig. 1B). Natural killer cells number was not modified by atRA (not shown). To more accurately characterize the lymphocyte population targeted by atRA, we performed double staining using markers specific for the following populations of T lymphocytes: Th lymphocytes ($CD5^+/CD4^+$), Tc/s lymphocytes ($CD5^+/CD8^+$) and $T_{TCR\gamma/\delta}$ lymphocytes ($CD5^+/TCR\gamma/\delta^+$). Supplementation with atRA increased the number of circulating Th lymphocytes ($CD5^+/CD4^+$) and Tc/s lymphocytes ($CD5^+/CD8^+$) but not T lymphocytes bearing the TCRγ/δ (Fig. 1C). Taken together, these results show that atRA induces a doubling of the number of circulating T cells and that these cells are mainly Th lymphocytes.

3.2. All-trans retinoic acid enhances IL-2 expression in the liver

T cells are activated by interaction between T cell receptor (TCR) and antigens displayed on the surface of antigen presenting cells (APCs), and this, together with a second costimulatory signal, triggers a cascade of events that culminates in IL-2 production which drives the cells to proliferate (Robey and Allison, 1995). We hypothesized that the effect of atRA on Th cells number was consecutive to an increase of IL-2 production by T cells. However, IL-2 concentration in the plasma was not modified by atRA supplementation (Fig. 2A). We reasoned that this discrepancy between enhanced T cells number and absence of increased plasma IL-2 concentration may result from a rapid consumption of IL-2 by T cells leading to T cell proliferation. We stained T cells with CD25, the IL-2 receptor α (IL-2Rα), because binding of IL-2 to its receptor induces increased IL-2Rα expression. Only a minority of T lymphocytes (between 2 and 3%) carried out CD25 (Fig. 2B), and this was not modified by atRA. To further our understanding of the relationship between atRA and IL-2, we next evaluated IL-2 mRNA expression in lymphoid and non-lymphoid organs by SQ RT-PCR. Although IL-2 mRNA was expressed in the spleen and in the kidney, this expression was not modified by atRA (Fig. 2C). Subsequently, we studied IL-2 mRNA expression in the liver since we previously reported that atRA increased the LPS-induced IFN-γ expression in this organ (Devaux et al., 2000). While below the detection threshold of the assay in control rats, IL-2 mRNA expression was readily detectable following atRA supplementation (Fig. 2C). This increase was accompanied by a three-fold up-regulation of IL-2 protein expression, as measured by ELISA of liver homogenates (Fig. 2D). These data suggest that the IL-2/IL-2 receptor pathway is not involved in atRA-mediated T cell proliferation in peripheral blood but IL-2 expression is up-regulated in the liver

Table 1
List of mRNAs of interest, expected size of amplified cDNAs, sequence of nucleotide primers and Genbank accession numbers

mRNA of interest and size of expected amplified cDNA	Primer sequences (sense and antisense 5′ → 3′)	Genbank accession number
β-Actin, 232 bp	GGACTTCGAGCAGGAGATGG GCACCGTGTTGGCGTAGAGG	U07786
IL-2, 456 bp	AGTCCTGCAAGCATGTACAG GCTTTGACAGATGGCTATCC	M22899
NOS2, 578 bp	GATCAATAACCTGAAGCCCG GCCCTTTTTTGCTCCATAGG	X76881
IFN-γ, 489 bp	AGCTCTGAGACAATGAGTG CCAGAATCAGCACCGACTC	AF010466
TLR4, 555 bp	CGCTTTCAGCTTTGCCTTCATTAC TGCTACTTCCTTGTGCCCTGTGAG	AF057025
CD14, 501 bp	TCACAATTCACTGCGGGATA GATCCCGAGGATCAAAATCA	AF087943
MD-2, 509 bp	TGGTCTTCCTGGCGAGTTT GCGGTGAATGATGGTGAAAT	AB018550
LBP, 500 bp	AAAGAGCAGGCACTGTGGTT CTCACTTGCACAGACCTCCA	NM017208

following atRA supplementation and can lead to enhanced IFN-γ production in this organ.

The main producers of IL-2 are T lymphocytes. We therefore hypothesized that the increased IL-2 expression in the liver of atRA-supplemented rats was consecutive to T cells infiltration in that organ. To test this, liver sections were subjected to haematoxylin and eosin staining. No infiltrated leukocytes were visible in liver sections from either control or atRA-supplemented animals (not shown). These data suggest that T cells are not responsible for enhanced IL-2 production in the liver of atRA-supplemented rats.

3.3. All-trans retinoic acid modulates the expression of several components of the LPS recognition system in the liver

Recognition of LPS by the innate immune system involves binding of LPS to the macrophage-expressed CD14 by LBP. Because LBP is produced by hepatocytes, a reservoir of Vitamin A in the liver, we hypothesized that atRA could modify LBP production. Supplementation with atRA induced a two-fold increase of LBP mRNA expression in the liver (Fig. 3). The LPS signaling pathway involves a complex formed by LPS, CD14, MD-2 and TLR4 together with other proteins, such as CD11c and CD18 integrins, CD55, HSP 70 and 90 among others (Triantafilou and Triantafilou, 2002). We investigated potential effects of atRA on some of the above-cited components of the LPS recognition system. Supplementation with atRA did not modify the expression of CD14, MD-2 and TLR4. While LBP and CD14 mRNA expression was up-regulated following LPS treatment, TLR4 mRNA expression was down-regulated. In addition, atRA increased the LPS-induced CD14 expression (Fig. 3). Therefore, these data show that supplementation with atRA induces a priming of the immune response to LPS through enhanced LBP expression.

3.4. A RARα agonist but not a pan-RXRs agonist mimicks the effects of atRA on LBP expression

We have previously reported that atRA enhances LPS-induced NOS2 activation through a RARα-dependent mechanism (Seguin-Devaux et al., 2002). Herein, we sought to investigate whether the atRA-mediated enhancement of LBP expression also involved a RARα-dependent pathway. To test this, rats were supplemented with the RARα agonist Ro 4060-55 or the pan-RXRs agonist Ro 2573-86 for 5 days. We observed that the RARα agonist but not the pan-RXRs agonist mimicked the effects of atRA on LBP mRNA expression in the liver (Fig. 4A) and LBP concentration in the plasma (Fig. 4B).

3.5. The effects of atRA supplementation on the IFN-γ and NOS2 pathways depend on the stimulus

The present data showing enhanced LBP expression following atRA supplementation suggest that the immune response to LPS may be stimulated by atRA and may explain the increased IFN-γ expression and NOS2 activation in LPS-treated rats (Devaux et al., 2000). To get some insight on the importance of the enhancement of LBP expression by atRA on host–pathogen interactions, atRA-supplemented rats were infected with the Gram-positive bacteria LM. Preliminary experiments were performed to document that the bacterial load was maximal in the liver and spleen 24 h following infection, and that this was correlated with IFN-γ and NOS2 expression in these organs (not shown). Thus, 24 h after infection with LM, the bacterial load was measured in the liver and spleen and was found similar between these two organs and between atRA-supplemented and vehicle-supplemented rats (Fig. 5A). In striking contrast with our previous results obtained when rats were challenged with LPS, NOS2 and IFN-γ mRNA expression in the liver and spleen were not significantly different between rats from atRA

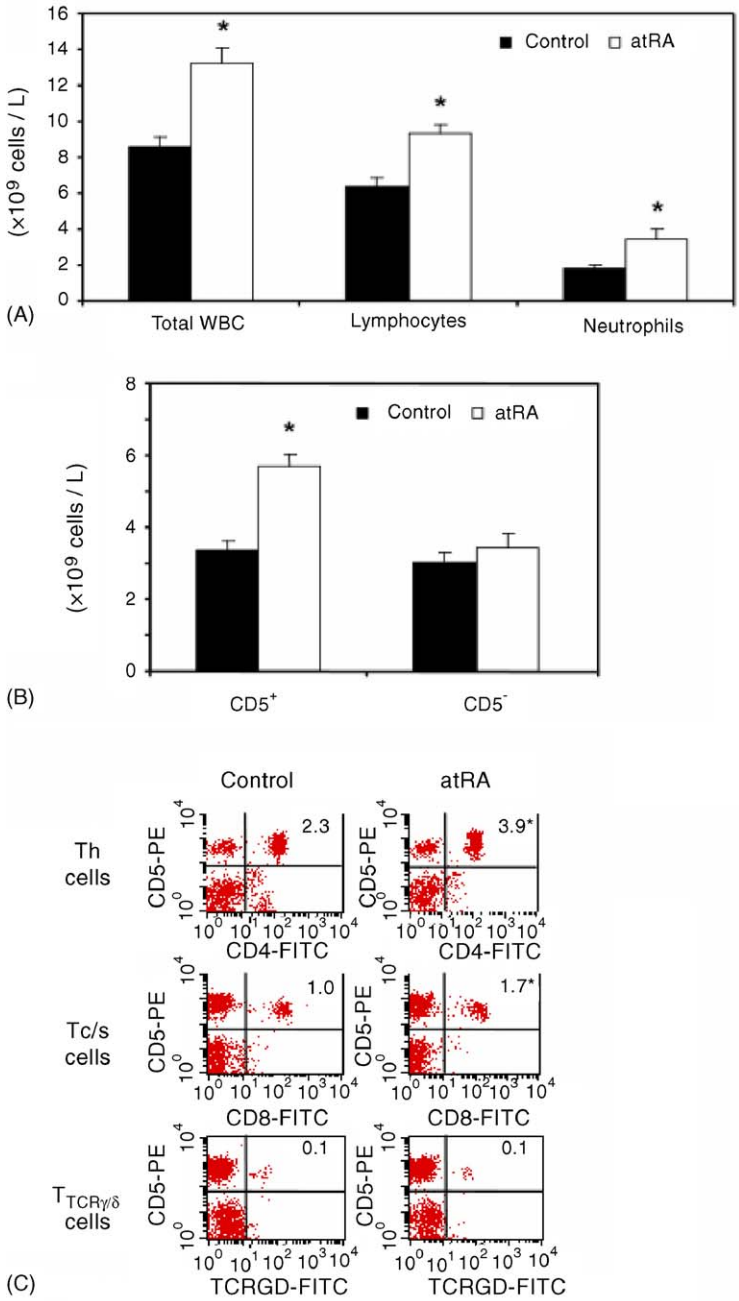

Fig. 1. Supplementation with atRA enhances peripheral blood T lymphocytes number. Rats received 10 mg/kg atRA (atRA) or vehicle (control) daily for 5 days. (A) Peripheral blood count revealed that atRA supplementation enhanced the numbers of circulating leukocytes, lymphocytes and neutrophils. (B) Flow cytometry analysis showed that atRA increased the number of circulating $CD5^+$ cells but not the number of $CD5^-$ cells. Values are mean ± S.E.M. (C) atRA increased the numbers of circulating Th ($CD5^+/CD4^+$) and Tc/s ($CD5^+/CD8^+$) lymphocytes. γ/δ T cells ($CD5^+/TCR\gamma/\delta^+$) number was not modified by atRA. Cell numbers ($\times 10^9$ cells/L blood) indicated on the upper right corner of each dot-plot were deduced from the peripheral blood count and the percentage of double positive cells. Results are expressed as means ± S.E.M. $^*p < 0.05$ vs. control.

and atRA + LM groups. Rather than an increase of NOS2 and IFN-γ expression, atRA had a tendency to decrease the LM-triggered activation of these two components of innate immunity (Fig. 5B and C). These data show that the immune response of the host after supplementation with atRA depends on the stimulus and suggest that the increased LBP expression can explain the deleterious effect of atRA supplementation in LPS-stimulated rats.

4. Discussion

The results of the present study show that supplementation with atRA in rats induces a “priming” of the host immune response characterized by increased number of circulating T lymphocytes and increased expression of one component of the host LPS recognition system, the LPS binding protein. The enhanced LBP expression is probably responsible for the

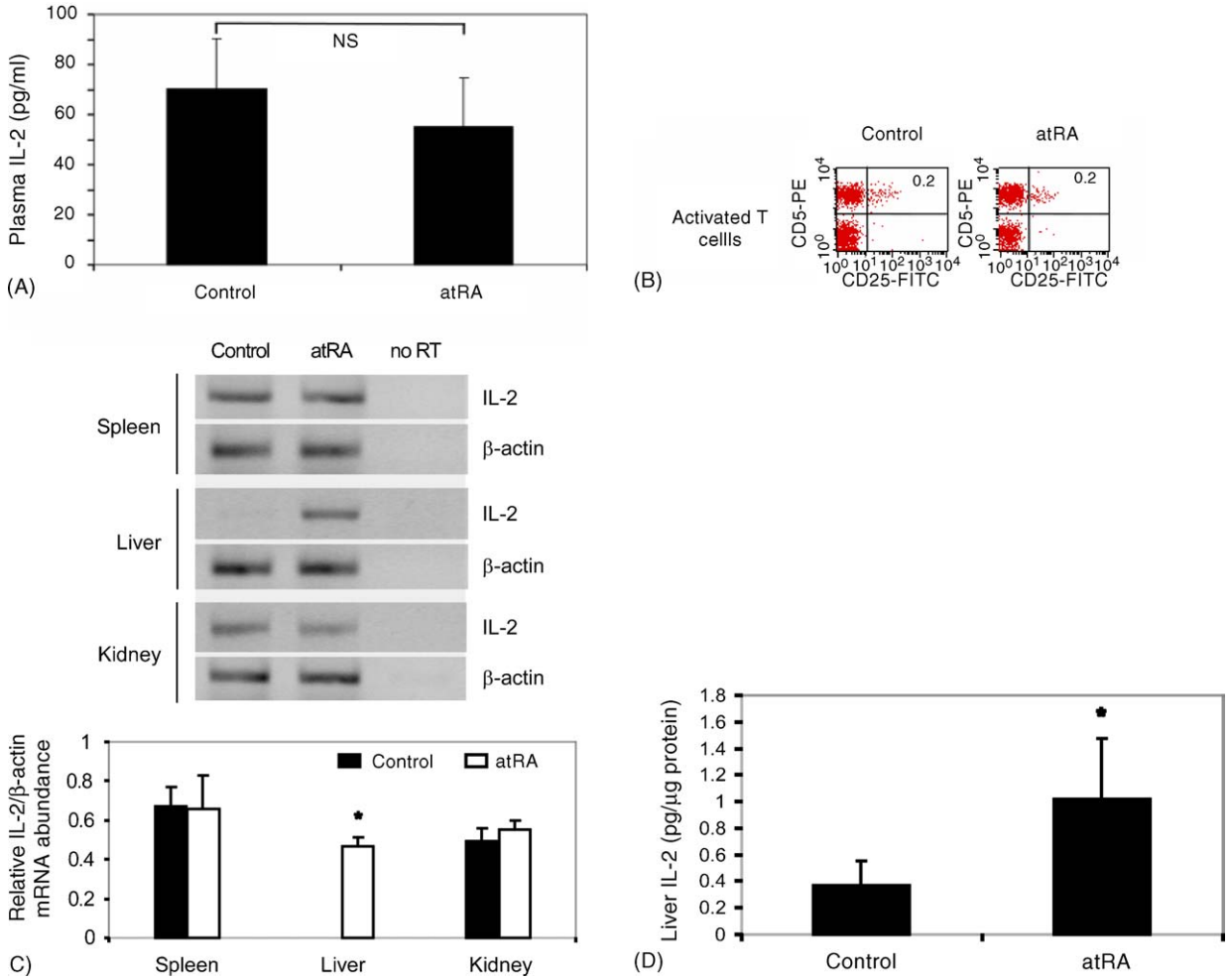

Fig. 2. Supplementation with atRA increases IL-2 production in the liver. Rats received 10 mg/kg atRA (atRA) or vehicle (control) daily for 5 days. (A) atRA did not significantly modify IL-2 plasma concentration. Plasma IL-2 concentration was measured by ELISA and is expressed as mean ± S.E.M. (pg/mL). NS: not significant. (B) The number of $CD5^+/CD25^+$ T cells in the peripheral blood was similar between control and atRA-supplemented rats. Cell numbers ($\times 10^9$ cells/L blood) indicated on the upper right corner of each dot-plot were deduced from the peripheral blood count and the percentage of double positive cells. (C) While undetectable under basal conditions, IL-2 mRNA expression in the liver was induced following atRA supplementation as determined by SQ RT-PCR. The absence of amplification product when reverse transcriptase RT was omitted (−RT) attested for the absence of genomic DNA contamination in the extracted RNA. A representative profile is shown (upper panel). Densitometric analysis was performed and IL-2 mRNA expression in the liver is expressed as the mean of relative IL-2/β-actin mRNA abundance ± S.E.M. (lower panel). ND: not detected. (D) IL-2 protein expression in the liver was also increased by atRA, as evaluated by ELISA. Results are mean ± S.E.M. (pg/µg protein). $^*p < 0.05$ vs. control.

divergent effects of atRA supplementation on NOS2 and IFN-γ pathways between LPS- and *L. monocytogenes*-challenged rats.

In preliminary experiments, we observed that a single dose of 10 mg/kg atRA did not show any effect on IL-2, LBP expression and circulating T cells number (not shown). This is in agreement with the study of van Pelt and de Rooij (1991) where replicate doses of atRA are necessary to have biological effects. Potential toxicity associated with the protocol of atRA supplementation used in this study, i.e. five daily injections of 10 mg/kg atRA, has been discussed elsewhere (Devaux et al., 2000). A toxic effect of atRA is unlikely since replicate doses of atRA had no effect on rat behavior, mortality or on visual inspection of the peritoneal cavity. In addition, there were no signs of hypervitaminosis A, such as weight loss, hair loss or skin scaling. These observations are in agreement with Look et al. (1995) who showed toxic effects of atRA in mice only after 7 days of supplementation with 100 mg/kg/day. The rat model of non-lethal endotoxemia has been previously characterized by members of our group (Boczkowski et al., 1996; Sambe et al., 1998). Preliminary experiments revealed that the dose of 4 mg/kg LPS and the time course of 6 h following LPS administration used in this study were a good compromise between a non-lethal protocol and a significant effect, particularly on LBP expression.

The first observation consistent with a priming of the host immune response concerns the increased number of circulating T lymphocytes. Our in vivo results suggest that supplementation with atRA is responsible for increased number of circulating T lymphocytes, mostly $CD4^+$ T lymphocytes. This result is in agreement with a previous study showing that repletion of Vitamin A-deficient rats with atRA is able to restore the number of blood lymphocytes (Zhao and Ross, 1995). Due to the known effects of Vitamin A on hematopoesis, atRA may have a

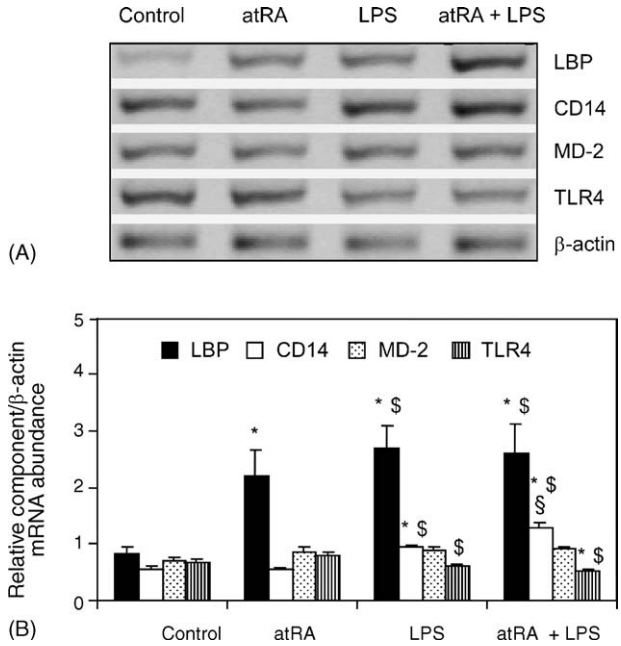

Fig. 3. Effects of atRA and LPS on mRNA expression of the different components of the LPS recognition system in the liver. Rats received 10 mg/kg atRA (atRA) or vehicle (control) daily for 5 days and were challenged with LPS (LPS) or vehicle (atRA + LPS). LBP, CD14, MD-2 and TLR4 mRNA expression was evaluated by SQ RT-PCR. atRA induced a two-fold increase of liver LBP mRNA expression and LPS enhanced LBP, CD14 and TLR4 expression. LPS challenge resulted in higher CD14 expression in atRA-supplemented than in non-supplemented rats. (A) A representative RT-PCR profile is shown. (B) Densitometric analysis. The optical density values of the PCR products of interest were divided by that of the β-actin product. Results are expressed as mean ± S.E.M. $^{*,\$,\S}p < 0.05$ vs. control, atRA and LPS groups, respectively.

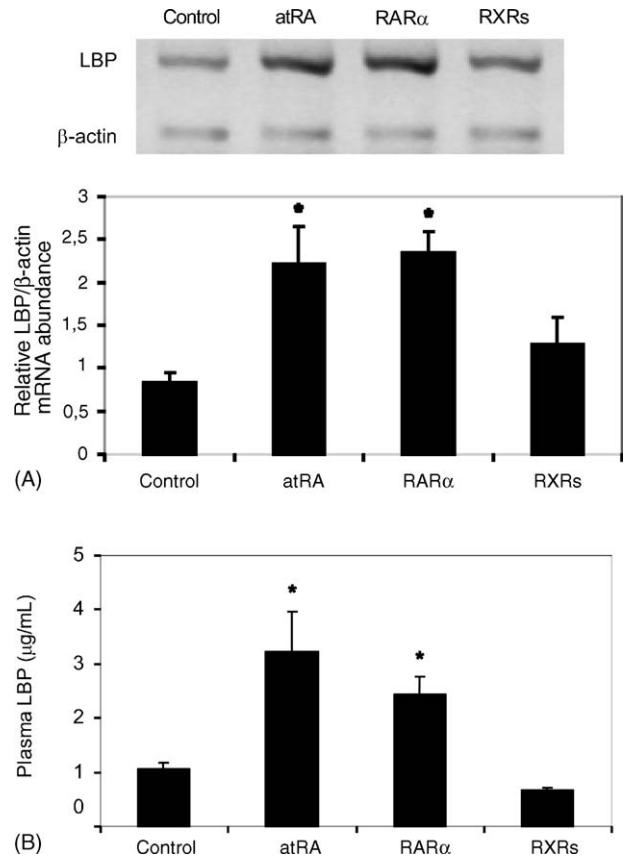

Fig. 4. Effects of atRA on LBP expression are mimicked by a RARα but not a pan-RXRs agonist. Rats were supplemented with 10 mg/kg of atRA, the RARα agonist Ro-4060-55 or the pan-RXRs agonist Ro 2573-86 for 5 days. (A) atRA and the RARα but not the pan-RXRs agonist induced a more than two-fold increase of liver LBP mRNA expression. A representative RT-PCR profile of rat liver LBP mRNA expression is shown (upper panel). The optical density value of the LBP PCR product was divided by that of the β-actin product (lower panel). (B) atRA and the RARα but not the pan-RXRs agonist induced a three-fold increase of plasma LBP concentration. LBP plasma concentration was determined by ELISA and is expressed in μg/mL. Results are mean ± S.E.M. $^{*}p < 0.05$ vs. control group.

non-specific effect on the entire population of leukocytes in this experimental model. However, a global effect of atRA on leukocytes is unlikely since the numbers of circulating B lymphocytes and monocytes were not modified by atRA supplementation.

To investigate the mechanism involved in the increased T lymphocytes number in the peripheral blood of atRA-supplemented rats, we investigated the IL-2 pathway. IL-2 concentration in the plasma was not different between control and atRA-supplemented rats. However, IL-2 secretion in T cells is difficult to measure due to rapid internalization of secreted IL-2 once it binds its receptor (Fujii et al., 1986). Therefore, we investigated CD25 expression, the IL-2Rα, by flow cytometry and we found that IL-2Rα expression was not modified by atRA supplementation. Assuming that IL-2Rα expression is increased consequently to IL-2 binding, our results indicate that the IL-2/IL-2Rα pathway is not involved in atRA-mediated T cell proliferation. Ertesvag et al. (2002) reported that the proliferative effect of atRA on cultured human T lymphocytes was associated with increased secretion of IL-2 but that IL-2 secretion by itself cannot fully explain the proliferative effects of atRA. In addition, these authors failed to detect any increase of IL-2Rα expression by atRA (Ertesvag et al., 2002). These in vitro results are consistent with our in vivo observations showing that atRA supplementation did not modify IL-2Rα expression.

The design of our experiments does not allow us to assert that the increased circulating T lymphocytes number is due to increased proliferation or inhibition of apoptosis. Both stimulation of the cell cycle (Ertesvag et al., 2002) and inhibition of apoptosis (Toth et al., 2001; Engedal et al., 2004) have been shown to mediate the proliferative effects of atRA on activated T lymphocytes in vitro. A correlation between the abilities of atRA to inhibit apoptosis of T lymphocytes and to increase IL-2 production has been reported (Engedal et al., 2004) and RARα binding by the synthetic agonist CD336 inhibited apoptosis induced by anti-CD3 or a specific Ag (Szegezdi et al., 2003).

The relationship between atRA and IL-2 has been investigated in several organs in the present study. We chose the spleen as a lymphoid organ and the kidney as a non-lymphoid organ. We also investigated IL-2 expression in the liver, since hepatocytes are known to be a reservoir of Vitamin A and because we previously observed that the main effect of atRA on IFN-γ expression was located in the liver (Devaux et al., 2000). Although, atRA enhanced IL-2 expression in the liver, atRA had no effect on IL-2 expression either in the spleen or in the kidney. In addition,

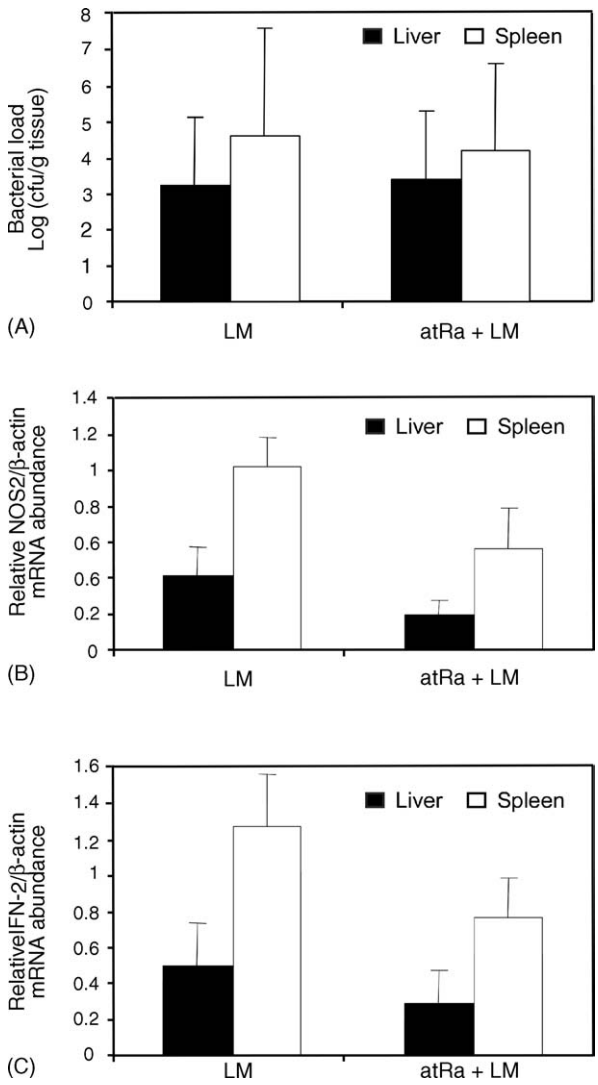

Fig. 5. atRA does not modulate IFN-γ and NOS2 expression in *Listeria monocytogenes*-infected rats. Rats were supplemented with 10 mg/kg atRA (atRA + LM group, $n = 5$) or vehicle (LM group, $n = 5$) daily for 5 days before infection with 10^9 CFU LM i.p., and were sacrificed 24 h after. (A) LM burden in the liver and spleen was similar between atRA-supplemented and vehicle-supplemented rats. Organs were weighed, homogenized and plated onto blood-gelose. CFU were counted after 24 h and bacterial load is expressed in log (CFU/g tissue). Although not statistically significantly ($p > 0.05$), atRA had a tendency to decrease NOS2 (B) and IFN-γ (C) mRNA expression in the liver and spleen. IFN-γ and NOS2 mRNA expression was estimated by SQ RT-PCR. Densitometric analysis was performed and the optical density values of the IFN-γ and NOS2 PCR products were divided by that of the β-actin product. Results are expressed as mean ± S.E.M.

liver sections from atRA-supplemented rats did not display any infiltrated cells, whereas examination of liver sections from LPS-treated rats revealed a strong infiltration of inflammatory cells. These results suggest that lymphocytes are not implicated in the induction of IL-2 expression by atRA in the liver, but rather a cell type specific to the liver. IL-2 expression has been shown in hepatocytes from transplanted rat liver and was correlated with IFN-γ expression (Lord et al., 2001). Together with the existing literature, our results suggest that hepatocytes may be responsible for IL-2 expression following atRA supplementation.

The second result consistent with a "priming" of the host immune response after atRA supplementation involves the increased liver mRNA and plasma concentration of LBP. The molecular mechanisms of LPS recognition by the host have been reviewed (Heine and Lien, 2003). In the plasma, LPS is recruited to soluble CD14 by LBP and the complex LPS/LBP binds the macrophage membrane-bound CD14 and the TLR4/MD-2 complex. This latter complex serves as the LPS signaling receptor which activates a cascade of molecular pathways leading to the activation of the transcription factor NF-κB regulating cytokine production by immune cells (Heine and Lien, 2003). Whereas CD14, MD-2 and TLR4 liver mRNA expression was not modified by atRA, a significant increase of LBP liver mRNA expression and plasma concentration was induced by atRA. The critical importance of LBP in the host LPS recognition system is documented. LBP knocked-out mice are unable to fight infection by LPS or Gram-negative bacteria (Jack et al., 1997) and LBP greatly enhances the ability of LPS to bind to CD14 (Wright et al., 1990). Thus, LBP can be considered as a limiting factor of the host response to LPS. Our results demonstrate for the first time that atRA supplementation in vivo increased LBP mRNA and protein expression. This effect is specific of LBP since the expression of the other components of LPS recognition system (CD14, MD-2 and TLR4) was not modified by atRA.

The mechanisms of the atRA-mediated increase in LBP mRNA and protein expression are not known. Retinoids, such as atRA, modulate gene transcription by direct binding to their cognate receptors and subsequently to RARE or by protein–protein interactions with different transcription factors, such as the AP-1 complex, SP-1, NF-κB and STAT3. The rat LBP promoter contains several putative consensus sequences, such as SP-1, NF-κB, GRE, STAT-3, IRF2, GATA, C/EBP, c-jun and c-myc (Kirschning et al., 1997). A regulatory element that was found to be crucial for transcription and activation of the LBP gene is the APRE/STAT3 binding site (Schumann et al., 1996). Interestingly, Yan et al. (2002) reported that STAT3 showed interactions with RAR enhanceosome proteins, such as RARα, in pulmonary epithelial cells. Using the AliBaba2 program for predicting transcription factor binding sites (http://www.gene-regulation.com/pub/programs/alibaba2/index.html), we found that the LBP gene promoter has a putative RA response element (RARE). Our results suggest that atRA enhancement of LBP transcription involves RARα. However, whether RARα interacts with STAT3 or directly binds to a putative RARE in the LBP promoter in our experimental model is not known.

Consistently with a previous report (Payne et al., 1992), our results show that LPS increases CD14 mRNA expression in rat liver in vivo. Additionally, they suggest for the first time that, whereas atRA had no effect alone, it enhanced the LPS-induced increase of CD14 mRNA expression in the rat liver. There was no major effects of RA supplementation on MD-2, TLR2 and TLR4, whereas their mRNA expression before and after stimulation with LPS is consistent with previous studies (Akashi et al., 2000; Matsuguchi et al., 2000; Matsumura et al., 2000).

Using different pro-inflammatory/infectious stimuli, we show that the effect of atRA on LBP expression is critical, since it provides an explanation why atRA enhances NOS2 and IFN-

γ expression when the infectious stimulus is LPS, a component of the bacterial wall of Gram-negative bacteria (Devaux et al., 2000), but not when the stimulus is the Gram-positive bacteria LM. Also, our results provide an explanation for the opposite effects of atRA on NOS2 pathway in several monocellular cultures, such as smooth muscle cells, vascular endothelial cells or cardiac myocytes (Hirokawa et al., 1994; Mehta et al., 1994; Datta and Lianos, 1999; Datta et al., 2001; Motomura et al., 1997; Grosjean et al., 2001) and in vivo (Devaux et al., 2000; Seguin-Devaux et al., 2002). Effectively, we show here in vivo that atRA supplementation activates at least two other cell types, lymphocytes and hepatocytes. Importantly, our results showing enhanced production of LBP by atRA highlight the critical role of hepatocytes in the modulation of the immune response.

In conclusion, we show here that supplementation with atRA triggers a "priming" of the host immune system which affects cellular and non-cellular components of the immune responses. These results contribute to characterize the modulatory effects of Vitamin A on host–pathogen interactions and specifically the situations where Vitamin A supplementation can be beneficial for host survival.

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